	Model (Red means implemented in RMark)	Code	Example
1	Live Recaptures (CJS)	CJS	?dipper; ?example.data
2	Dead Recoveries	Recovery	?brownie
3	Both Live and Dead Encounters -- Burnham	Burnham	?Burnham
4	Known Fate	Known	?Blackduck
5	Closed Population Estimation	Closed	?edwards.eberhardt
6	BTO Dead Recoveries and Unknown Ringings	BTO	
7	Robust Design with Closed Population Estimation	Robust	?robust
8	Both Live and Dead Encounters -- Barker	Barker	
9	Multi-state with Live Recaptures	Multistrata	
10	Brownie et al. Dead Recoveries	Brownie	?brownie
11	Jolly-Seber Lambda -- Burnham	Jolly	
12	Huggins Closed Population Estimation	Huggins	?edwards.eberhardt
13	Robust Design with Huggins' Estimator	RDHuggins	?robust
14	Pradel Recruitment Only	Pradel	
15	Pradel Survival and Seniority	PradSen	
16	Pradel Survival and Lambda	PradLambda	
17	Pradel Survival and Recruitment	PradRec	
18	Barker Live and Dead with Closed Robust Design	RDBarker	
19	POPAN	POPAN	?dipper
20	Virtual Population Analysis (VPA)	VPA	
21	Multi-state with Live and Dead Encounters	MSLiveDead	
22	Closed Captures with Heterogeneity	HetClosed	?edwards.eberhardt
23	Full Closed Captures with Heterogeneity	FullHet	?edwards.eberhardt
24	Nest Success	Nest	?killdeer; ?mallard
25	Huggins' Closed Captures with Heterogeneity	HugHet	?edwards.eberhardt
26	Huggins' Full Closed Captures with Heterogeneity	HugFullHet	?edwards.eberhardt
27	Occupancy Estimation with Detection < 1	Occupancy	?salamander; ?weta
28	RD Occupancy Estimation with psi, epsilon.	RDOccupPE	?RDSalamander
29	RD Occupancy Estimation with psi, gamma.	RDOccupPG	?RDOccupancy; ?RDSalamander
30	RD Occupancy Estimation with psi(1), gamma, epsilon.	RDOccupEG	?RDSalamander

31	Link-Barker Jolly-Seber	LinkBarker	
32	Open Robust Design Multi-state	ORDMS	
33	Closed Robust Design Multi-state	CRDMS	?crdms
34	Huggins' Closed Robust Design Multi-state	HCRDMS	
35	Heterogeneity Closed Robust Design Multi-state	HetRDMS	
36	Full Heterogeneity Closed Robust Design Multi-state	FHetRDMS	
37	Huggins' Het. Closed Robust Design Multi-state	HHetRDMS	
38	Huggins' Full Het. Closed Robust Design Multi-state	HFHetRDMS	
39	Robust Design with Heterogeneity Estimator	RDHet	?robust
40	Robust Design with Full Heterogeneity Estimator	RDFullHet	
41	Robust Design with Huggins' Het. Estimator	RDHHet	
42	Robust Design with Huggins' Full Het. Estimator	RDHFHet	
43	Barker Live and Dead with Huggins' Robust Design	RDBarkHug	
44	Barker Live and Dead with Heterogeneity Robust Design	RDBarkHet	
45	Barker Live and Dead with Full Het. Robust Design	RDBarkFHet	
46	Barker Live and Dead with Huggins' Het. Robust Design	RDBarkHHet	
47	Barker Live and Dead with Huggins' Full Het. Robust Design	RDBarkHFHet	
48	Lukacs Young Survival from Marked Adults	LYSMA	
49	Closed Population Estimation with mis-identification	IDClosed	
50	Huggins Closed Populations with mis-identification	IDHuggins	
51	Closed Captures with Heterogeneity and mis-identification	IDHetClosed	
52	Full Closed Captures with Het. and mis-identification	IDFullHet	
53	Huggins' Closed Captures with Het. and mis-identification	IDHugHet	
54	Huggins' Full Closed Captures with Het. and mis-identification	IDHugFullHet	
55	Robust Design Closed Population Estimation with mis-identification	RDIDClosed	
56	Robust Design Huggins' Closed Populations with mis-identification	RDIDHuggins	
57	Robust Design Closed Captures with Heterogeneity and mis-identification	RDIDHetClosed	
58	Robust Design Full Closed Captures with Het. and mis-identification	RDIDFullHet	
59	Robust Design Huggins' Closed Captures with Het. and mis-identification	RDIDHugHet	
60	Robust Design Huggins' Full Closed Captures with Het. and mis-identification	RDIDHugFullHet	
61	Robust Design Multi-state Closed Population Estimation with mis-identification	CMSIDClosed	
62	Robust Design Multi-state Huggins' Closed Populations with mis-identification	CMSIDHuggins	

63	Robust Design Multi-state Closed Captures with Heterogeneity and mis-identification	CMSIDHetClosed	
64	Robust Design Multi-state Full Closed Captures with Het. and mis-identification	CMSIDFullHet	
65	Robust Design Multi-state Huggins' Closed Captures with Het. and mis-identification	CMSIDHugHet	
66	Robust Design Multi-state Huggins' Full Closed Captures with Het. and mis-identification	CMSIDHugFullHet	
67	Robust Design Pradel Seniority Closed Population Estimation	RDPdGClosed	
68	Robust Design Pradel Seniority Huggins' Closed Populations	RDPdGHuggins	
69	Robust Design Pradel Seniority Closed Captures with Heterogeneity	RDPdGHet	
70	Robust Design Pradel Seniority Full Closed Captures with Het.	RDPdGFullHet	
71	Robust Design Pradel Seniority Huggins' Closed Captures with Het.	RDPdGHugHet	
72	Robust Design Pradel Seniority Huggins' Full Closed Captures with Het.	RDPdGHugFullHet	
73	Robust Design Pradel Seniority Closed Population Estimation with mis-identification	RDPdGIDClosed	
74	Robust Design Pradel Seniority Huggins' Closed Populations with mis-identification	RDPdGIDHuggins	
75	Robust Design Pradel Seniority Closed Captures with Heterogeneity and mis-identification	RDPdGIDHet	
76	Robust Design Pradel Seniority Full Closed Captures with Het. and mis-identification	RDPdGIDFullHet	
77	Robust Design Pradel Seniority Huggins' Closed Captures with Het. and mis-identification	RDPdGIDHugHet	
78	Robust Design Pradel Seniority Huggins' Full Closed Captures with Het. and mis-identification	RDPdGIDHugFullHet	
79	Robust Design Pradel Lambda Closed Population Estimation	RDPdLClosed	
80	Robust Design Pradel Lambda Huggins' Closed Populations	RDPdLHuggins	
81	Robust Design Pradel Lambda Closed Captures with Heterogeneity	RDPdLHet	
82	Robust Design Pradel Lambda Full Closed Captures with Het.	RDPdLFullHet	
83	Robust Design Pradel Lambda Huggins' Closed Captures with Het.	RDPdLHugHet	
84	Robust Design Pradel Lambda Huggins' Full Closed Captures with Het.	RDPdLHugFullHet	
85	Robust Design Pradel Lambda Closed Population Estimation with mis-identification	RDPdLIDClosed	
86	Robust Design Pradel Lambda Huggins' Closed Populations with mis-identification	RDPdLIDHuggins	
87	Robust Design Pradel Lambda Closed Captures with Heterogeneity and mis-identification	RDPdLIDHet	
88	Robust Design Pradel Lambda Full Closed Captures with Het. and mis-identification	RDPdLIDFullHet	
89	Robust Design Pradel Lambda Huggins' Closed Captures with Het. and mis-identification	RDPdLIDHugHet	
90	Robust Design Pradel Lambda Huggins' Full Closed Captures with Het. and mis-identification	RDPdLIDHugFullHet	
91	Robust Design Pradel Recruitment Closed Population Estimation	RDPdfClosed	
92	Robust Design Pradel Recruitment Huggins' Closed Populations	RDPdfHuggins	
93	Robust Design Pradel Recruitment Closed Captures with Heterogeneity	RDPdfHet	
94	Robust Design Pradel Recruitment Full Closed Captures with Het.	RDPdfFullHet	

95	Robust Design Pradel Recruitment Huggins' Closed Captures with Het.	RDPdfHugHet	
96	Robust Design Pradel Recruitment Huggins' Full Closed Captures with Het.	RDPdfHugFullHet	
97	Robust Design Pradel Recruitment Closed Population Estimation with mis-identification	RDPdfIDClosed	
98	Robust Design Pradel Recruitment Huggins' Closed Populations with mis-identification	RDPdfIDHuggins	
99	Robust Design Pradel Recruitment Closed Captures with Heterogeneity and mis-identification	RDPdfIDHet	
100	Robust Design Pradel Recruitment Full Closed Captures with Het. and mis-identification	RDPdfIDFullHet	
101	Robust Design Pradel Recruitment Huggins' Closed Captures with Het. and mis-identification	RDPdfIDHugHet	
102	Robust Design Pradel Recruitment Huggins' Full Closed Captures with Het. and mis-identification	RDPdfIDHugFullHet	
103	Open Robust Design Pradel Multi-state	ORDPdMS	
104	Huggins Closed Robust Design Multi-state with State Probabilities	CRDMSOHug	
105	Huggins Heterogeneity Closed Robust Design Multi-state with State Probabilities	CRDMSOHet	
106	Huggins Full Heterogeneity Closed Robust Design Multi-state with State Probabilities	CRDMSOFHet	
107	Occupancy Heterogeneity Estimation with Detection < 1	OccupHet	?salamander
108	RD Occupancy Heterogeneity Estimation with psi, epsilon	RDOccupHetPE	
109	RD Occupancy Heterogeneity Estimation with psi, gamma	RDOccupHetPG	
110	RD Occupancy Heterogeneity Estimation with psi(1), gamma, epsilon	RDOccupHetEG	
111	Occupancy Estimation Royle/Nichols Poisson Abundance	OccupRNPoisson	?Donovan.7
112	Occupancy Estimation Royle/Nichols Negative Binomial Abundance	OccupRNNegBin	?Donovan.7
113	Two species Occupancy Estimation	2SpecOccup	
114	Logit-Normal Mark Resight	LogitNormalMR	?LogitNormalMR
115	Poisson Mark Resight with Robust Design	PoissonMR	?PoissonMR, ?Poisson_twoMR
116	Multiple-State Occupancy Estimation	MSOccupancy	?NicholsMSOccupancy
117	Occupancy Estimation Royle Poisson Counts	OccupRPoisson	?Donovan.8
118	Occupancy Estimation Royle Negative Binomial Counts	OccupRNegBin	?Donovan.8
119	Open Robust Design Multi-state with State Probabilities	ORDMSState	
120	Immigration-Emigration Logit-Normal Mark Resight	IELogitNormalMR	?IELogitNormalMR
121	Robust Design Multi-state Closed with Mis-classification	RDMSMisClass	
122	Robust Design Multi-state Closed with 2 Mis-classifications	RDMS2MisClass	
123	Multi-scale occupancy estimation	MultScalOcc	?larksparrow
124	Robust Design Multiple-State Occupancy Estimation Conditional Binomial	RDMSOccRepro	
125	Robust Design Multiple-State Occupancy Estimation General	RDMSOccupancy	
126	Robust Design Multi-state Open with Mis-classification	RDMSOpenMisClass	

127	Density estimation with Huggins p and c	Densitypc	?Density
128	Density estimation with Huggins heterogeneity pi and p	DensityHet	?Density
129	Density estimation with Huggins full heterogeneity pi, p and c	DensityFHet	?Density
130	Cormack-Jolly-Seber model with Pledger mixtures	CJSMixture	
131	Pradel Survival and Seniority with Pledger mixtures	PradSenMix	
132	Pradel Survival and Lambda with Pledger mixtures	PradLambdaMix	
133	Pradel Survival and Recruitment with Pledger mixtures	PradelRecMix	
134	Link-Barker Survival and Recruitment with Pledger mixtures	LinkBarkMix	
135	Cormack-Jolly-Seber model with Random Effects	CJSRandom	
136	Link-Barker Survival and Recruitment with Random Effects	LinkBarkRan	
137	Two species Conditional Occupancy Estimation	2SpecConOccup	
138	Burnham Live and Dead Encounters with Random Effects	BurnhamLDRE	
139	Pledger Mixture Dead Recoveries (Seber)	PMDead	?brownie
140	Random Effects Dead Recoveries (Seber)	REDead	?brownie
141	Robust Design Two species Gamma Epsilon Conditional Occupancy Estimation	RD2SpGEConOcc	
142	Robust Design Multi-state Open with State Uncertainty and Seasonal Effects	RDMSOpenMCSeas	
143	Occupancy with correlated detections	OccClus	
144	Occupancy with relaxed closure	OccRelClos	
145	Huggins' p and c with Random Effects	HugginsRE	
146	Robust Design with Huggins' p and c with Random Effects	RDHugginsRE	
147	Closed Robust Design Multi-state Huggins' p and c with Random Effects	HRECRDMS	
148	Robust Design Pradel Seniority Huggins' p and c with Random Effects	RDPDGHUGRE	
149	Robust Design Pradel Lambda Huggins' p and c with Random Effects	RDPDLHUGRE	
150	Robust Design Pradel f Huggins' p and c with Random Effects	RDPDFHUGRE	
151	Occupancy Estimation with Detection < 1 and Random Effects	OccupancyRE	
152	Robust Design Occupancy Estimation with psi, epsilon and Random Effects	RDOccupREPE	
153	Robust Design Occupancy Estimation with psi, gamma and Random Effects	RDOccupREPG	
154	Robust Design Occupancy Estimation with psi(1), gamma, epsilon and Random Effects	RDOccupREEG	
155	Known Fate with Random Effects	KnownRE	
156	Occupancy Estimation with false positive identifications	OccupancyFP	
157	Robust Design Occupancy Estimation with psi(1), gamma, epsilon and false positives.	RDOccupFPEG	

158	UnIdLogitNormalMR: Unidentified Marks Logit-Normal Mark Resight	UnIdLogitNormalMR	
159	UnIdIELogitNormalMR: Unidentified Marks Immigration-Emigration Logit-Normal Mark Resight	UnIdIELogitNormalMR	
160	UnIdPoissonMR: Unidentified Marks Poisson Mark Resight with Robust Design	UnIdPoissonMR	
161	Density estimation with Huggins' p and c with Random Effects	DensityRanpc	?Density
162	Barker Live and Dead with Robust Design Huggins' p and c with Random Effects	RDBarkHRE	
163	Robust Design Pradel Full Likelihood p and c with lambda from ratio of N's	RDPdNClosed	
164	Robust Design Pradel Huggins' p and c with lambda from ratio of N's	RDPdNHuggins	
165	Robust Design Pradel Full Likelihood Heterogeneity pi and p with lambda from ratio of N's	RDPdNHet	
166	Robust Design Pradel Full Likelihood Heterogeneity pi, p, and c with lambda from ratio of N's	RDPdNFullHet	
167	Robust Design Pradel Huggins' Heterogeneity pi and p with lambda from ratio of N's	RDPdNHugHet	
168	Robust Design Pradel Huggins' Heterogeneity pi, p, and c with lambda from ratio of N's	RDPdNHugFullHet	
169	Robust Design Pradel Huggins' p and c with Random Effects with lambda from ratio of N's	RDPdNHUGRE	
170	Robust Design Occupancy Estimation with relaxed closure	RDOCCURELCLOS	
171	Zero-inflated Unidentified Marks Poisson Mark Resight with Robust Design within primaries	ZiUnIdPoissonMRwithin	
172	Poisson Mark Resight with Robust Design across primaries	PoissonMRacross	
173	Unidentified Marks Poisson Mark Resight with Robust Design across primaries	UnIdPoissonMRacross	
174	Zero-inflated Unidentified Marks Poisson Mark Resight with Robust Design across primaries	ZiUnIdPoissonMRacross	
175	Robust Design Multi-scale occupancy estimation	RDMultScalOcc	?RDMultScalOcc
176	Barker Multi-state Joint Live and Dead Encounters and Resightings	MSBarker	
177	Multi-species Occupancy Model	NSpeciesOcc	?NSpeciesOcc
178	Hidden Markov Model	HidMarkov	
179	Barker Multi-state Joint Live and Dead Encounters and Resightings with multiple uncertain events	MSUncBarker	
180	Barker Multi-state Joint Live and Dead Encounters and Resightings with multiple uncertain events and transitons first	MSUnc2Barker	
181	Multi-state with Live Recaptures with 2 Random Effects	Multistate2RE	
182	Multi-state with Live Recaptures with 3 Random Effects	Multistate3RE	
183	Multi-state Jolly-Seber	MSJollySeber	
184	Robust Design Multi-state Open with State Uncertainty and Age and Seasonal Effects	RDMSOpenMCSeas2	
185	Hidden Markov Multi-state Jolly-Seber	HMMMSJollySeber	

Parameters

1	Phi	p								
2	S	r								
3	S	p	r	F						
4	S									
5	p	c	f0							
6	S									
7	S	Gamma''	Gamma'	p	c	f0				
8	S	p	r	R	R'	F	F'			
9	S	p	Psi							
10	S	f								
11	Phi	p	Lambda	N						
12	p	c								
13	S	Gamma''	Gamma'	p	c					
14	Gamma	p								
15	Phi	p	Gamma							
16	Phi	p	Lambda							
17	Phi	p	f							
18	S	r	R	R'	a''	a'	F	p	c	f0
19	Phi	p	pent	N						
20	M	F								
21	S	p	Psi	r						
22	pi	p	f0							
23	pi	p	c	f0						
24	S									
25	pi	p								
26	pi	p	c							
27	p	Psi								
28	Psi	Epsilon	p							
29	Psi	Gamma	p							
30	Psi	Epsilon	Gamma	p						
31	Phi	p	f							
32	S	Psi	pent	Phi	p					

33	S	Psi	p	c	f0						
34	S	Psi	p	c							
35	S	Psi	pi	p	f0						
36	S	Psi	pi	p	c	f0					
37	S	Psi	pi	p							
38	S	Psi	pi	p	c						
39	S	Gamma''	Gamma'	pi	p	f0					
40	S	Gamma''	Gamma'	pi	p	c	f0				
41	S	Gamma''	Gamma'	pi	p						
42	S	Gamma''	Gamma'	pi	p	c					
43	S	r	R	R'	a''	a'	F	p	c		
44	S	r	R	R'	a''	a'	F	pi	p	f0	
45	S	r	R	R'	a''	a'	F	pi	p	c	f0
46	S	r	R	R'	a''	a'	F	pi	p		
47	S	r	R	R'	a''	a'	F	pi	p	c	
48	Phi	p									
49	p	c	alpha	f0							
50	p	c	alpha								
51	pi	p	alpha	f0							
52	pi	p	c	alpha	f0						
53	pi	p	alpha								
54	pi	p	c	alpha							
55	S	Gamma''	Gamma'	p	c	alpha	f0				
56	S	Gamma''	Gamma'	p	c	alpha					
57	S	Gamma''	Gamma'	pi	p	alpha	f0				
58	S	Gamma''	Gamma'	pi	p	c	alpha	f0			
59	S	Gamma''	Gamma'	pi	p	alpha					
60	S	Gamma''	Gamma'	pi	p	c	alpha				
61	S	Psi	p	c	alpha	f0					
62	S	Psi	p	c	alpha						
63	S	Psi	pi	p	alpha	f0					
64	S	Psi	pi	p	c	alpha	f0				
65	S	Psi	pi	p	alpha						

66	S	Psi	pi	p	c	alpha	
67	Phi	Gamma	p	c	f0		
68	Phi	Gamma	p	c			
69	Phi	Gamma	pi	p	f0		
70	Phi	Gamma	pi	p	c	f0	
71	Phi	Gamma	pi	p			
72	Phi	Gamma	pi	p	c		
73	Phi	Gamma	p	c	alpha	f0	
74	Phi	Gamma	p	c	alpha		
75	Phi	Gamma	pi	p	alpha	f0	
76	Phi	Gamma	pi	p	c	alpha	f0
77	Phi	Gamma	pi	p	alpha		
78	Phi	Gamma	pi	p	c	alpha	
79	Phi	Lambda	p	c	f0		
80	Phi	Lambda	p	c			
81	Phi	Lambda	pi	p	f0		
82	Phi	Lambda	pi	p	c	f0	
83	Phi	Lambda	pi	p			
84	Phi	Lambda	pi	p	c		
85	Phi	Lambda	p	c	alpha	f0	
86	Phi	Lambda	p	c	alpha		
87	Phi	Lambda	pi	p	alpha	f0	
88	Phi	Lambda	pi	p	c	alpha	f0
89	Phi	Lambda	pi	p	alpha		
90	Phi	Lambda	pi	p	c	alpha	
91	Phi	f	p	c	f0		
92	Phi	f	p	c			
93	Phi	f	pi	p	f0		
94	Phi	f	pi	p	c	f0	
95	Phi	f	pi	p			
96	Phi	f	pi	p	c		
97	Phi	f	p	c	alpha	f0	
98	Phi	f	p	c	alpha		

99	Phi	f	pi	p	alpha	f0		
100	Phi	f	pi	p	c	alpha	f0	
101	Phi	f	pi	p	alpha			
102	Phi	f	pi	p	c	alpha		
103	S	Psi	Gamma	pent	Phi	p		
104	S	Psi	Omega	p	c			
105	S	Psi	Omega	pi	p			
106	S	Psi	Omega	pi	p	c		
107	pi	p	Psi					
108	Psi	Epsilon	pi	p				
109	Psi	Gamma	pi	p				
110	Psi	Epsilon	Gamma	pi	p			
111	r	Lambda						
112	r	Lambda	VarAdd					
113	PsiAB	PsiA	PsiB	pA	pB	rAB	rAb	raB
114	p	sigma	N					
115	alpha	sigma	U	Phi	Gamma''	Gamma'		
116	Psi1	Psi2	p1	p2	Delta			
117	r	Lambda						
118	r	Lambda	VarAdd					
119	S	Psi	Omega	pent	Phi	p		
120	p	sigma	Nbar	alpha	Nstar			
121	S	Psi	pi	Omega	p	Delta		
122	S	Psi	pi	Omega	p	Delta		
123	Psi	Theta	p					
124	Phi0	Psi	R	p	Delta			
125	Phi0	Psi	p					
126	S	Psi	pi	Omega	p	Delta	pent	Phi
127	p	c	ptilde					
128	pi	p	ptilde					
129	pi	p	c	ptilde				
130	pi	Phi	p					
131	Phi	pi	p	Gamma				

132	Phi	pi	p	Lambda												
133	Phi	pi	p	f												
134	Phi	pi	p	f												
135	sigmaphi	Phi	sigmap	p												
136	sigmaphi	Phi	sigmap	p	sigmaf	f										
137	PsiA	PsiBA	PsiBa	pA	pB	rA	rBA	rBa								
138	sigmaS	S	sigmap	p	sigmar	r	sigmaF	F								
139	pi	S	r													
140	sigmaS	S	sigmar	r												
141	PsiA	PsiBA	PsiBa	GammaA	GammaB	GammaAB	GammaBA	EpsilonA	EpsilonB	EpsilonAB	EpsilonBA	pA	pB	rA	rBA	rBa
142	S	Psi	pi	Omega	p	Delta	pent	Phi	alpha	c						
143	Psi	Theta	Theta'	p												
144	Psi	pent	d	p												
145	sigmap	p	c													
146	S	Gamma''	Gamma'	sigmap	p	c										
147	S	Psi	sigmap	p	c											
148	Phi	Gamma	sigmap	p	c											
149	Phi	Lambda	sigmap	p	c											
150	Phi	f	sigmap	p	c											
151	sigmap	p	Psi													
152	Psi	Epsilon	sigmap	p												
153	Psi	Gamma	sigmap	p												
154	Psi	Epsilon	Gamma	sigmap	p											
155	sigmaS	S														
156	p10	p11	b	Psi												
157	Psi	Epsilon	Gamma	p10	p11	b										
158	p	sigma	N	r												
159	p	sigma	Nbar	alpha	Nstar	r										
160	alpha	sigma	U	r	Phi	Gamma''	Gamma'									
161	sigmap	p	c	ptilde												
162	S	r	R	R'	a''	a'	F	sigmap	p	c						
163	Phi	p	c	f0												

164	Phi	p	c								
165	Phi	pi	p	f0							
166	Phi	pi	p	c	f0						
167	Phi	pi	p								
168	Phi	pi	p	c							
169	Phi	sigmap	p	c							
170	Psi	Epsilon	Gamma	pent	d	p					
171	alpha	sigma	U	r	w	g	Phi	Gamma''	Gamma'		
172	alpha	sigma	U	Phi	Gamma''	Gamma'					
173	alpha	sigma	U	r	Phi	Gamma''	Gamma'				
174	alpha	sigma	U	r	w	g	Phi	Gamma''	Gamma'		
175	Psi	Epsilon	Gamma	Theta	p						
176	S	Psi	p	R	R'	r	Delta	rho	rho'	b	pi
177	f	p									
178	S	p	Psi	pi	Delta						
179	S	Psi	p	R	R'	r	Delta	rho	rho'	b	pi
180	S	Psi	p	R	R'	r	Delta	rho	rho'	b	pi
181	sigmaS	S	sigmap	p	Psi						
182	sigmaS	S	sigmap	p	sigmaPsi	Psi					
183	S	p	Psi	pent	pi						
184	S	Psi	pi	Omega	p	Delta	pent	Phi	alpha	c	
185	S	p	Psi	pent	pi	Delta					